A · ANCHOR

GENERAL ARRANGEMENT — CHARGING STATION

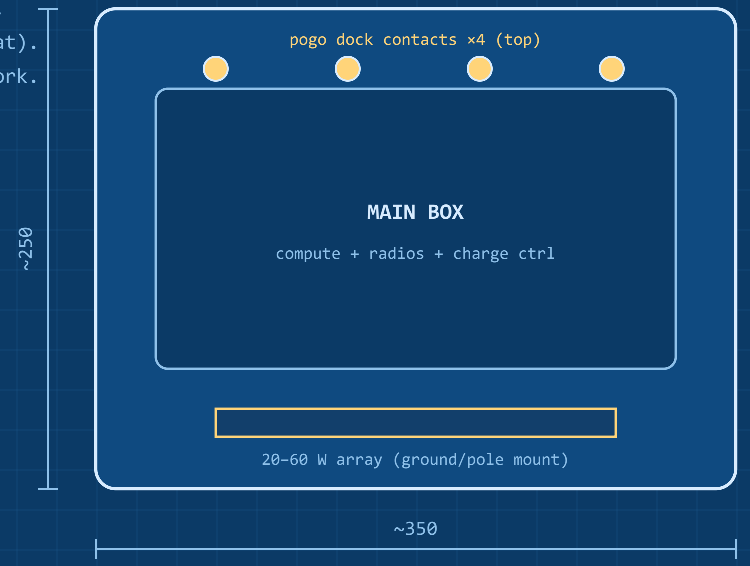
RENEWABLE STATION: SWARM DOCK + RELAY + BACKHAUL

AF-A-101
REV A · 2026-11-12
PROTOTYPE

WHAT THIS NODE IS

- The fixed, renewable-powered infrastructure the swarm needs — a station where Miters top off and the mesh gets stronger.
- Sited at the community hub: band office, fire lookout, well site, school — wherever people already gather.
- Three jobs in one box: swarm charging dock, long-range LoRa relay, optional backhaul gateway (LTE/Wi-Fi/sat).
- The only node whose job is charging OTHER nodes — it turns the swarm from visitors into a persistent network.

ENVELOPE — STATION


pogo dock contacts ×4 (top)
MAIN BOX
compute + radios + charge ctrl
20–60 W array (ground/pole mount)
~250
~350

box ~350×250×400 mm incl. battery bay · + separate array

WHAT THE ANCHOR DOES

Function	Detail
Swarm dock	4 pogo charge positions on top — Miters land, charge, rebroadcast
Long-range relay	SX1262 at height + bigger antenna — extends mesh over ridges/valleys
Gateway	LTE/Wi-Fi/sat backhaul when available — the mesh's door to the world
Anchor sensor	Air temp/RH/VOC + wind — the fixed reference point for swarm data
Status beacon	E-Ink or LED shows mesh health — infrastructure you can read

WHY ANCHORS MATTER

- Swarm energy: docked Miters top off → perpetual patrol without battery swaps.
- Range: elevated relay doubles effective mesh diameter vs. ground nodes.
- Trust: the station is the visible, maintainable heart of the local mesh.
- Revenue anchor: gateways are what ISPs/communities pay for — the swarm rides free.

ARCTICCHAIN · RURAL NODE FLEET

PROPRIETARY & CONFIDENTIAL — EMEBET INC.	SOLE PROPRIETARY OWNER	**DWG AF-A-101**
DRAWN: Surafel Negussie — Planner / Maker	**Emebet Inc. · emebetos.ca** registered Canadian company · Edmonton, AB	REV A SCALE NTS · SHEET 1/3

DESIGNED FOR RURAL & REMOTE CANADA — -45..+70 °C · IP65 · SOLAR-FIRST · COMMUNITY-OWNED

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MECHANICAL & POWER SYSTEM

ENCLOSURE, ARRAY, BATTERY, DOCK

AF-A-102

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PROTOTYPE

ENCLOSURE & PLACEMENT

Parameter	Value	Parameter	Value
[illegible]ound mount)			
[illegible]/fiberglass tool-required			
[illegible]h access compartment			
[illegible]edor pole			
[illegible]eor trailer			

POWER SYSTEM — SIZED FOR WINTER

20–60 W ARRAY oversized for Dec sun → **MPPT 20–30 A** LiFePO4 profile → **LiFePO4 50–100 Ah** 12.8 V ≈ 640–1280 Wh → **BMS → 5 V/12 V rails** station + dock power

Autonomy target: 30–90 days no-sun (smaller load than it looks — LoRa is µW-class, compute sleeps).

SWARM DOCK — ELECTRICAL

- 4 pogo-pin positions on top lid: VBUS 5 V + GND + charge-detect line.
- Each dock: ~5 V / 500 mA charge path through CN3791-class controller per position.
- Mite bottom has 2 pogo pads + alignment notch — lands blind, charges, rebroadcasts.
- Dock schedule: rotating charge slots so no unit starves; mesh reports dock occupancy.

ENCLOSURE LAYOUT

- Top: dock contacts + antenna bulkheads (LoRa whip, LTE whip, GPS patch).
- Mid: compute bay (CM5-class) + radio bay + charge controllers.
- Bottom: sealed battery compartment (LiFePO4, vented, BMS external-access).
- Side: cable entry glands, status LED bar, optional small E-Ink status window.

SITE EXAMPLES

- Fire lookout tower base — elevated relay + swarm dock for patrol units.
- Band office / community hall — gateway + public Totem co-located.
- Well site / pump house — telemetry + mesh node where workers already go.
- School — charging dock doubles as device-charging station for students.
- Trailhead / boat launch — seasonal swarm staging point.

SAFETY & CODE NOTES

- LiFePO4: safer chemistry than Li-ion; BMS mandatory; vent battery compartment.
- Electrical: licensed electrician for permanent install; GFCI on AC backup input.
- Lightning: DC-side surge protection on solar + antenna ports (field replaceable).
- Wildlife: no food/scent; box is electronics-only — still use community-standard mounting.

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SCALE NTS · SHEET 2/3

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ELECTRICAL — BOM, BACKHAUL & DOCK

PARTS + CONNECTIVITY OPTIONS

AF-A-103

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PROTOTYPE

BILL OF MATERIALS (PER STATION)

#	Part	Supplier class	Approx CAD	Qty
	[illegible] (gateway compute)			
	[illegible] ×2 (node band + backbone)			
	[illegible] bridge OR sat terminal			
	[illegible] panel + MPPT 20–30 A			
	[illegible] 100 Ah + BMS			
	[illegible] pole/frame			
	[illegible] charge controllers			
	[illegible] ×2, LTE whip, GPS patch			

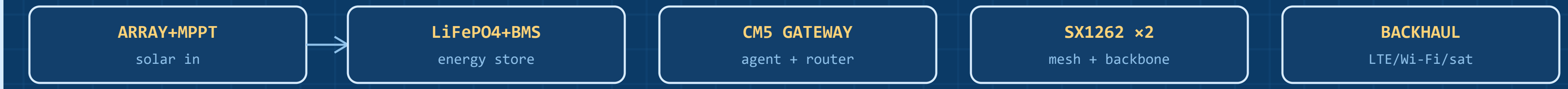
BACKHAUL OPTIONS (HIERARCHY)

Option	Cost	When
[illegible] — still fully functional as relay/dock		
[illegible] — most rural sites		
[illegible] case required		
[illegible] future own-sat path		
[illegible]		

DOCK PROTOCOL

- Mite lands on pogo pads → charge-detect line pulled low.
- Dock controller ID broadcasts slot assignment over LoRa.
- Charge until >80% or dock timeout; then release + 'charged' beacon.
- Rotating priority: lowest-battery docked unit gets next slot.
- All dock events signed + logged — swarm energy is mesh data.

SYSTEM BLOCK



Dock controllers hang off I2C/GPIO; antenna ports on box top; all rails fused.

FIELD ACCEPTANCE

- 4 Miters dock, charge, undock on schedule — 72 h unattended soak test.
- Mesh throughput: ≥10 signed msgs/min relayed through Anchor at 5 km spacing.
- Backhaul failover: LTE drops → mesh store-forward continues → auto-reconnect.
- Winter test: -30 °C ambient, panel snow-covered 3 days → station still docks+relays.

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